

Trainings ▾

Worksheet

Note : The following procedure is **only** to be used for diesel engines with fixed time injectors (no STC or HVT).

The engine firing order is 1-5-3-6-2-4.

The cylinders are numbered from the front gear cover end of the engine.

Two crankshaft revolutions are required to adjust all of the valves and the injectors.

One pair of valves and one injector are adjusted at each pulley index mark before rotating the engine to the next index mark.

The valves and the injectors on the same cylinder are **not** adjusted at the same mark.

Each cylinder has three rocker levers. The lever nearest to the front of the engine is the exhaust lever.

The following table can be used for recording values.

Inspection Report for Valve Lash and Injector Travel									
	Work Sequence	VS Mark and Valves to be Set	Intake	Exhaust	Reset (Yes/No)	Crosshead (OK/Reset)	Cylinder Number Injector	Injector Travel	Reset (Yes/No)
		C-1					5		
		A-5					3		
		B-3					6		
		C-6					2		
		A-2					4		

Customer:

Location:

Unit Number:

Hours:

Date:

Job Number:

Engine Model Number:

Engine Serial Number:

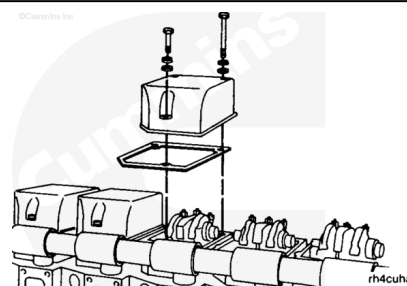
Type of inspection (failure, routine, complaint):

Inspection notes:

Note : This page can be copied for convenience.

Adjust

Remove the rocker lever cover and related components. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/18/18-003-011-om-ind.html>) in Section A.



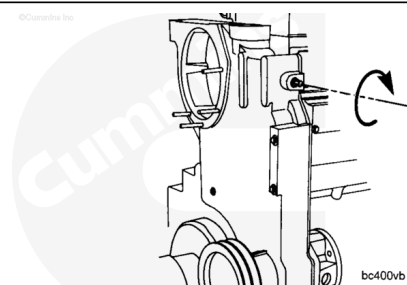
The valve and injector adjustment marks are found on the outside diameter of the accessory drive pulley or timing disk.

Some older engines have different index marks than A, B, and C.

On older engines:

- 1-6 VS is the same as A
- 2-5 VS is the same as B
- 3-4 VS is the same as C.

VS represents the valve set. Ignore any 1-6 TC mark during adjustment.



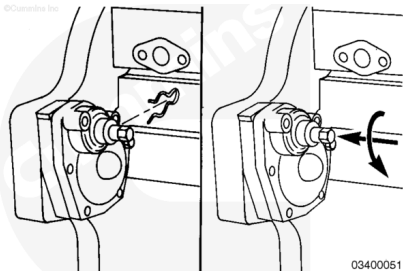
The barring device shaft turns approximately two revolutions before the engine begins to turn. The device will **not** turn the engine opposite the direction of normal rotation.

Push the shaft in and turn the barring device until the A mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.

On engines with a two-piece front cover:

- Remove the clip.
- Push the shaft in to engage the gears.
- Rotate the device shaft **counterclockwise** to turn the engine in the direction of normal rotation.

The alignment mark is also on the boss for the accessory drive seal.



The valves will be adjusted on the cylinder that has all of the valves closed. Use the table to determine the cylinders to check for valve position.

If Valve Set Mark Is:	Check Valve Position On:
A	1,6
B	2,5
C	3,4

If the rocker lever assemblies have been removed, use this step to determine the cylinder to set.

All of the adjusting screws **must** be loose on all cylinders and the push rod **must** remain in alignment.

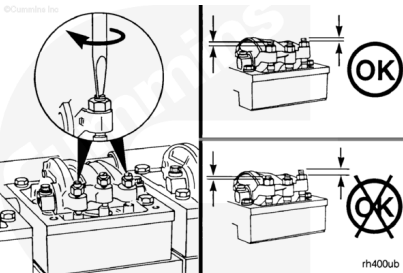
Perform this step on both of the cylinders to be checked.

Hold the rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods.

Turn the locknuts until they touch the rocker levers.

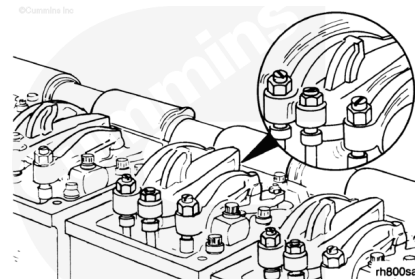
Compare the height of the adjusting screws above the locknut. The cylinder the adjusting screws are the same height are ready for valve adjustment.

The push rods will be the same height above the top of the rocker lever housing onto the cylinder ready for valve adjustment.



If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question.

Set the valves on the cylinder where both levers feel loose.



Use the chart to determine the injector that is ready to adjust.

Adjustment can begin on any valve set mark.

In the example, make sure the A mark is aligned and the adjusting screw height indicates the valves on cylinder number two are closed (ready to set). The chart shows the injector on cylinder number four is ready to adjust.

After the adjustment, bar the engine to the B set mark.

Adjust the valves on cylinder number four and the injector on cylinder number one.

	Closed On	V	I
A	1	5	3
B	5	3	6
C	3	6	2
Ⓐ	ⓖ	Ⓣ	Ⓓ
B	2	4	1
C	4	1	5

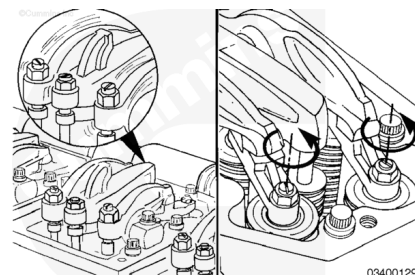
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Crossheads

Crosshead adjustment **must always** be made before attempting to adjust the valves.

Adjust the crossheads on the cylinder that has both valves closed.

Loosen the crosshead adjusting screw lock nuts on the intake and exhaust valve crossheads.

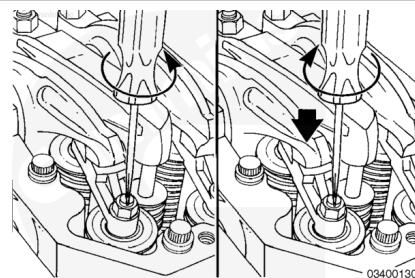


Use the following procedure to adjust both the intake and exhaust crossheads.

Turn the adjusting screw out at least one turn.

Hold the crosshead down against its mating valve stems.

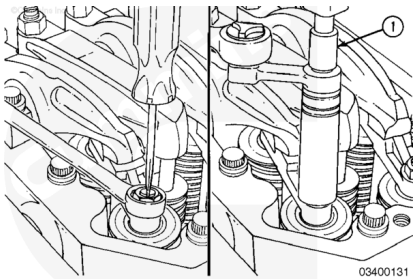
Turn the adjusting screw in until it touches the top of the valve stem but does **not** raise the crosshead.



The adjusting screw **must not** turn when the lock nut is tightened to its torque value.

Hold the adjusting screw in position.

Tighten the lock nut. The values are presented with and without the use of the torque wrench adapter (1), Part Number ST-669.



Torque Value:

Without Jacobs® Brake (With Adapter)
1. 35 n•m [25 ft-lb]

Torque Value:

Without Jacobs® Brake (Without Adapter)
1. 40 n•m [30 ft-lb]

Torque Value:

With Jacobs® Brake (With Adapter)
1. 40 n•m [30 ft-lb]

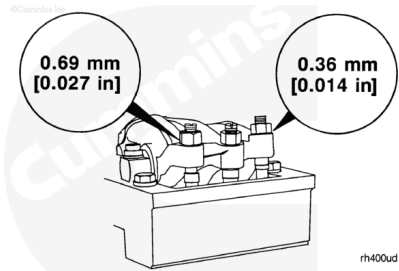
Torque Value:

With Jacobs® Brake (Without Adapter)
1. 45 n•m [33 ft-lb]

Valves

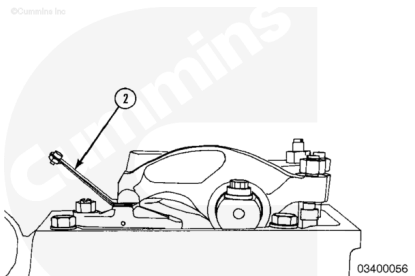
Use the table below to for valve adjustment initial set.

Valve Adjustment (Initial Set)	
Reference Point	Valve
A	Exhaust 0.69 mm [0.027 in]
B	Intake 0.36 mm [0.014 in]



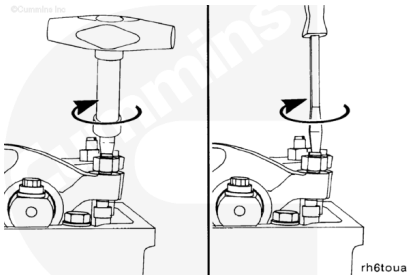
Select a feeler gauge for the correct valve lash specification.
Use service tool, Part Number 3163171 (intake) or Part Number 3163172 (exhaust).

Insert the gauge (2) between the rocker lever and the crosshead.



There are two different methods for establishing valve lash clearance. Either method can be used; however, the torque wrench method has proven to be the most consistent.

- The torque wrench method uses an inch pound torque wrench, Part Number 3376592. The adjusting screw is tightened to 0.68 N•m [6 in-lb].
- The feel method uses a screw driver and the adjusting screw is turned **only** until the lever touches the feeler



gauge.

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the locknut.

Torque Value:

With Torque Wrench Adapter, Part Number ST-669

1. 45 n•m [33 ft-lb]

Torque Value:

Without Adapter

1. 60 n•m [44 ft-lb]

The feeler gauge **must** slide backward and forward with **only** a slight drag.

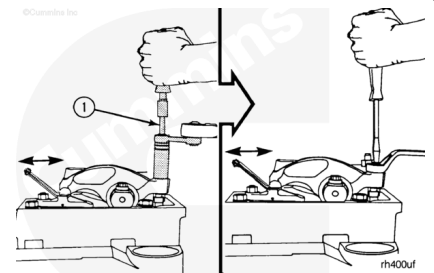
Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

Adjust the slave cylinder clearance on the Jacobs Brakes.

Refer to Procedure 020-999

(/qs3/pubsys2/xml/en/procedures/18/18-020-999.html).

Repeat the adjustment process until the clearance is correct on both the intake and exhaust valves.

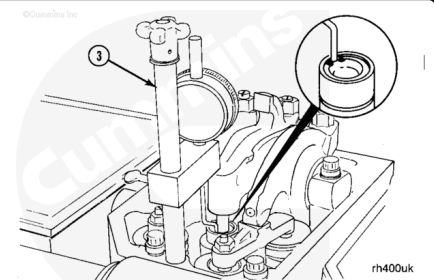


Injectors

Assemble the parts of the injector and valve adjustment kit (3), Part Number 3375004, or equivalent. Install the adjustment kit on the cylinder to be adjusted as illustrated in the graphic.

Adjust the indicator so the tip is touching the top of the injector plunger.

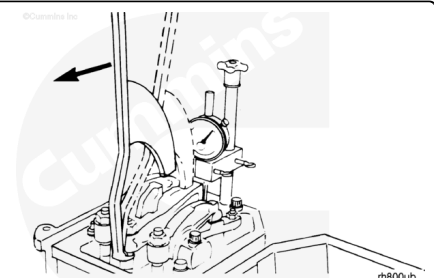
Lower the indicator 12.7 mm [0.500 in] to allow for travel. Lock the indicator support to the post.



⚠ WARNING ⚠

The injector plunger is under spring tension. Do not allow the tool to slip. Personal injury can result.

⚠ CAUTION ⚠



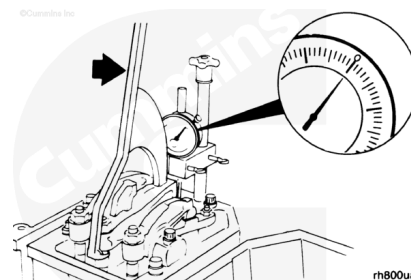
Prevent damage to the indicator by allowing the lever to return slowly.

Use a rocker lever actuator, Part Number 3376869, or equivalent, to depress the rocker lever until the injector bottoms two or three times.

This will remove fuel and oil from the cup.

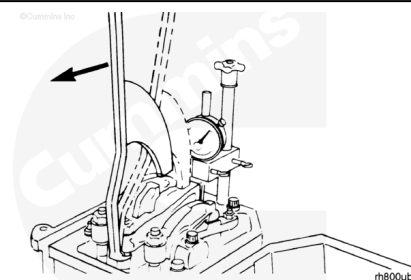
Hold the lever with the injector plunger firmly bottomed in the cup. Set the indicator to zero.

Raise and lower the lever a few time to confirm the zero.

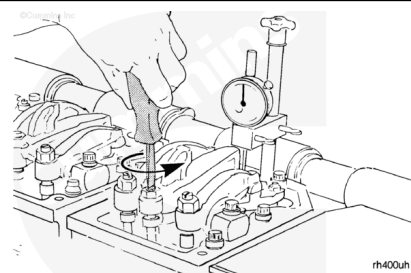


Slowly release the lever and observe the travel of the gauge.

Press down or tap lightly on the adjusting screw to confirm the reading.



Turn the adjusting screw until the indicator reads the specified travel.



Injector Travel Specifications	
Model	Specification
All fixed time	7.72 mm [0.304 in]

Recheck limits are intended to be used on an engine already in service. It is **not** intended as a tolerance to be used when adjusting.

Recheck Limits
7.67 to 7.77 mm [0.302 to 0.306 in]

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut.

Torque Value:

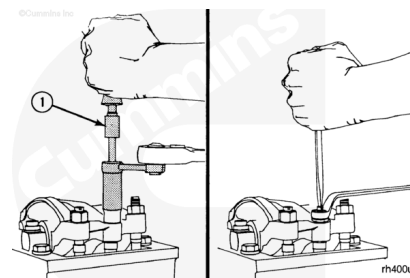
With Torque Wrench Adapter, Part Number ST-669

1. 45 n•m [33 ft-lb]

Torque Value:

Without Adapter

1. 60 n•m [44 ft-lb]

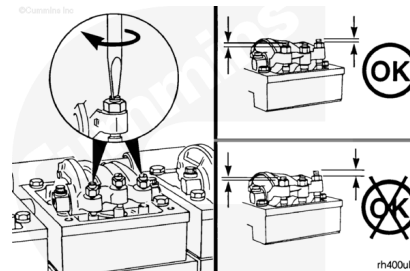


⚠ WARNING ⚠

The injector plunger is under spring tension. Do not allow the tool to slip. Personal injury can result.

⚠ CAUTION ⚠

Prevent damage to the indicator by allowing the lever to return slowly.

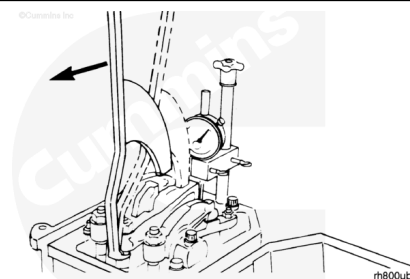


Check the injector adjustment. Use the rocker lever actuator.
Bottom the injector plunger.

Confirm zero on the indicator.

Allow the rocker lever to return slowly.

Check the injector setting. Repeat the adjustment process if it is **not** within specification.

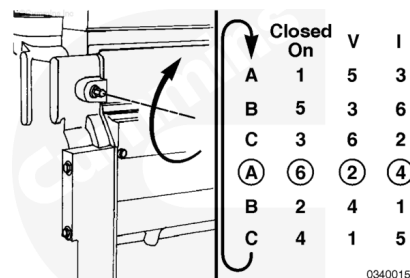


Rotate the engine.

Align the next mark.

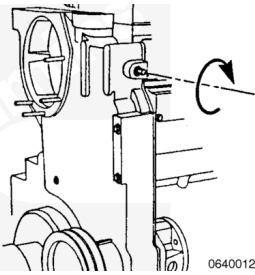
Adjust the appropriate valves and injectors.

Repeat the process to adjust all of the valves and the injectors correctly.



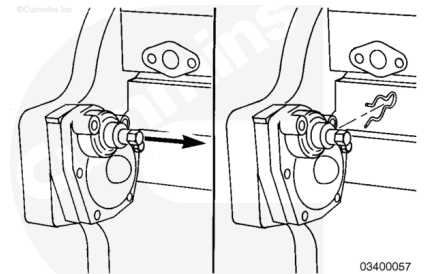
Turn the barring device in a **clockwise** direction to disengage the barring mechanism (worm gear).

If the worm gear remains engaged during the engine start, the engine rotation will disengage the parts without damage.



On engines with a two piece front cover:

- Pull the shaft out to disengage the gears.
- Install the clip.



Install the rocker lever cover and all related components. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/18/18-003-011-om-ind.html>) in Section A.

